

Lecture 3: Coarse Graining as a Probability-Learning Problem

1. Why Coarse Grain?

Atomistic molecular simulations retain a large number of degrees of freedom. For many problems, however, the detailed motion of every atom is not required.

Instead, we may define a reduced set of coarse-grained variables,

$$\mathbf{R} = M(\mathbf{x}), \quad (1)$$

where $\mathbf{x}$ denotes the microscopic atomistic coordinates and M is a coarse-graining map.

Examples include:

- replacing several atoms by a single interaction site;
- representing an amino acid by one or a few beads;
- describing a polymer by monomer-scale coordinates;
- retaining only collective coordinates relevant to a slow process.

The goal of coarse graining is not simply to reduce the number of particles. The coarse-grained model should reproduce selected properties of the underlying microscopic system, especially its equilibrium distribution.

The central question is therefore:

After microscopic degrees of freedom are removed, what effective potential should the remaining variables experience?

2. The Equilibrium Distribution of Coarse-Grained Variables

Suppose the microscopic system has potential energy

$$U_{AA}(\mathbf{x}) \quad (2)$$

and equilibrium probability distribution

$$p_{AA}(\mathbf{x}) = \frac{1}{Z_{AA}} e^{-\beta U_{AA}(\mathbf{x})}. \quad (3)$$

The equilibrium probability distribution of the coarse-grained variables is obtained by integrating over all microscopic configurations that map onto $\mathbf{R}$:

$$p_{CG}^*(\mathbf{R}) = \int d\mathbf{x} p_{AA}(\mathbf{x}) \delta[\mathbf{R} - M(\mathbf{x})]. \quad (4)$$

Substituting the Boltzmann distribution,

$$p_{CG}^*(\mathbf{R}) = \frac{1}{Z_{AA}} \int d\mathbf{x} e^{-\beta U_{AA}(\mathbf{x})} \delta[\mathbf{R} - M(\mathbf{x})]. \quad (5)$$

This distribution defines an exact effective coarse-grained potential.

3. Potential of Mean Force

The exact coarse-grained potential is defined through

$$p_{\text{CG}}^*(\mathbf{R}) \propto e^{-\beta U_{\text{PMF}}(\mathbf{R})}. \quad (6)$$

Therefore,

$$U_{\text{PMF}}(\mathbf{R}) = -k_{\text{B}}T \ln p_{\text{CG}}^*(\mathbf{R}) + C \quad (7)$$

or equivalently,

$$U_{\text{PMF}}(\mathbf{R}) = -k_{\text{B}}T \ln \left[\int d\mathbf{x} e^{-\beta U_{\text{AA}}(\mathbf{x})} \delta(\mathbf{R} - M(\mathbf{x})) \right] + C. \quad (8)$$

The coarse-grained potential is therefore a free-energy function.

Microscopic degrees of freedom do not simply disappear. Their energetic and entropic effects are encoded in U_{PMF} .

This gives an important conceptual relation:

$$\boxed{\text{eliminate degrees of freedom} \longrightarrow \text{generate effective free-energy interactions.}} \quad (9)$$

4. How Can We Determine a Coarse-Grained Potential?

The exact PMF is formally known,

$$U_{\text{PMF}} = -k_{\text{B}}T \ln p_{\text{CG}}^*, \quad (10)$$

but direct evaluation is usually impossible because p_{CG}^* is a high-dimensional distribution.

Instead, we construct a parameterized potential,

$$U_{\theta}(\mathbf{R}), \quad (11)$$

and determine θ from microscopic simulation data.

There are several possible strategies:

- reproduce mean forces;
- reproduce the full equilibrium probability distribution.

We first consider force matching.

5. Mean Forces

The exact CG force is the gradient of the potential of mean force,

$$\mathbf{F}_I^{\text{CG}}(\mathbf{R}) = -\nabla_I U_{\text{PMF}}(\mathbf{R}). \quad (12)$$

For a given microscopic configuration $\mathbf{x}$, one can compute an instantaneous projected force,

$$\mathbf{F}_I^{\text{inst}}(\mathbf{x}). \quad (13)$$

The exact mean force at a fixed CG configuration is the conditional average

$$\boxed{\mathbf{F}_I^{\text{CG}}(\mathbf{R}) = \langle \mathbf{F}_I^{\text{inst}}(\mathbf{x}) | M(\mathbf{x}) = \mathbf{R} \rangle}. \quad (14)$$

Thus individual microscopic configurations provide noisy samples of the underlying mean force.

6. Force Matching

Suppose the CG model predicts forces

$$\mathbf{F}_I^\theta(\mathbf{R}) = -\nabla_I U_\theta(\mathbf{R}). \quad (15)$$

Force matching determines the model parameters by minimizing

$$\mathcal{L}_{\text{FM}}(\theta) = \left\langle \sum_I \left| \mathbf{F}_I^{\text{inst}}(\mathbf{x}) - \mathbf{F}_I^\theta(M(\mathbf{x})) \right|^2 \right\rangle_{\text{AA}}. \quad (16)$$

Because mean-squared-error regression learns the conditional mean, the optimal force-matching model satisfies, within the chosen function class,

$$\mathbf{F}_I^\theta(\mathbf{R}) \approx \langle \mathbf{F}_I^{\text{inst}} | \mathbf{R} \rangle. \quad (17)$$

Thus force matching can be interpreted as a supervised learning problem:

$$\boxed{\mathbf{R} \longrightarrow \mathbf{F}} \quad (18)$$

where atomistic simulations provide the training labels.

7. From Force Learning to Probability Learning

Force matching focuses on reproducing mean forces.

But the fundamental equilibrium objective of coarse graining is more general: we want the CG model to reproduce the equilibrium distribution,

$$\boxed{p_\theta(\mathbf{R}) \approx p_{\text{CG}}^*(\mathbf{R})}. \quad (19)$$

The model distribution is

$$p_\theta(\mathbf{R}) = \frac{e^{-\beta U_\theta(\mathbf{R})}}{Z_\theta}, \quad (20)$$

where

$$Z_\theta = \int d\mathbf{R} e^{-\beta U_\theta(\mathbf{R})}. \quad (21)$$

Coarse graining can therefore be formulated as a probability-distribution learning problem.

This immediately establishes a connection with statistical inference and machine learning.

8. Maximum Likelihood

Suppose we have equilibrium coarse-grained configurations

$$\mathbf{R}_1, \dots, \mathbf{R}_N \sim p_{\text{CG}}^*. \quad (22)$$

We seek parameters θ that maximize the probability assigned by the model to these observations.

The likelihood is

$$\mathcal{L}(\theta) = \prod_{n=1}^N p_\theta(\mathbf{R}_n). \quad (23)$$

It is more convenient to maximize the log likelihood,

$$\ln \mathcal{L}(\theta) = \sum_{n=1}^N \ln p_{\theta}(\mathbf{R}_n). \quad (24)$$

Using

$$\ln p_{\theta}(\mathbf{R}) = -\beta U_{\theta}(\mathbf{R}) - \ln Z_{\theta}, \quad (25)$$

the negative log likelihood per configuration is

$$\mathcal{L}_{\text{NLL}} = \beta \langle U_{\theta}(\mathbf{R}) \rangle_{\text{target}} + \ln Z_{\theta}. \quad (26)$$

Thus fitting a Boltzmann model by maximum likelihood is directly equivalent to optimizing an effective potential.

9. Relative Entropy

Another way to quantify the difference between two probability distributions is the Kullback–Leibler divergence,

$$D_{\text{KL}}(p_{\text{target}} \| p_{\theta}) = \int d\mathbf{R} p_{\text{target}}(\mathbf{R}) \ln \frac{p_{\text{target}}(\mathbf{R})}{p_{\theta}(\mathbf{R})}. \quad (27)$$

In coarse graining this quantity is commonly referred to as the relative entropy.

Expanding,

$$D_{\text{KL}} = \int d\mathbf{R} p_{\text{target}} \ln p_{\text{target}} - \int d\mathbf{R} p_{\text{target}} \ln p_{\theta}. \quad (28)$$

The first term is independent of θ .

Since

$$\ln p_{\theta} = -\beta U_{\theta} - \ln Z_{\theta}, \quad (29)$$

we obtain

$$D_{\text{KL}} = \text{const.} + \beta \langle U_{\theta} \rangle_{\text{target}} + \ln Z_{\theta}. \quad (30)$$

Therefore,

relative-entropy minimization $\equiv$ maximum-likelihood learning of the Boltzmann distribution.

(31)

10. Gradient of the Relative Entropy

Differentiate the relative entropy with respect to θ :

$$\nabla_{\theta} D_{\text{KL}} = \beta \langle \nabla_{\theta} U_{\theta} \rangle_{\text{target}} + \nabla_{\theta} \ln Z_{\theta}. \quad (32)$$

Now,

$$\nabla_{\theta} \ln Z_{\theta} = \frac{1}{Z_{\theta}} \nabla_{\theta} \int d\mathbf{R} e^{-\beta U_{\theta}(\mathbf{R})} \quad (33)$$

$$= -\beta \frac{\int d\mathbf{R} e^{-\beta U_{\theta}} \nabla_{\theta} U_{\theta}}{Z_{\theta}} \quad (34)$$

$$= -\beta \langle \nabla_{\theta} U_{\theta} \rangle_{\theta}. \quad (35)$$

Thus,

$$\boxed{\nabla_{\theta} D_{\text{KL}} = \beta \left[\langle \nabla_{\theta} U_{\theta} \rangle_{\text{target}} - \langle \nabla_{\theta} U_{\theta} \rangle_{\theta} \right]} \quad (36)$$

This equation has an intuitive interpretation.

The first term describes what the target distribution samples.

The second term describes what the current CG model samples.

Optimization changes the potential until these two statistical averages agree.

11. The Computational Challenge of Maximum Likelihood

The gradient

$$\nabla_{\theta} D_{\text{KL}} = \beta \left[\langle \nabla_{\theta} U_{\theta} \rangle_{\text{target}} - \langle \nabla_{\theta} U_{\theta} \rangle_{\theta} \right] \quad (37)$$

contains two averages.

The target average can be estimated from the reference simulation data.

However,

$$\langle \nabla_{\theta} U_{\theta} \rangle_{\theta} \quad (38)$$

requires configurations sampled from the current model,

$$p_{\theta}(\mathbf{R}) \propto e^{-\beta U_{\theta}(\mathbf{R})}. \quad (39)$$

As θ changes during optimization, the equilibrium distribution changes as well.

A straightforward optimization therefore requires a repeated cycle:

$$\boxed{\theta \rightarrow \text{simulation under } U_{\theta} \rightarrow \text{model samples} \rightarrow \nabla_{\theta} D_{\text{KL}} \rightarrow \theta'}. \quad (40)$$

Repeated equilibrium simulations can become the dominant computational cost.

This motivates alternative approaches to probability learning.

12. Connection to Machine-Learning Force Fields

So far we have written the coarse-grained model abstractly as

$$U_{\theta}(\mathbf{R}). \quad (41)$$

Historically, U_{θ} was often represented using simple analytical functions such as pair potentials, bond potentials, and angle potentials.

Modern machine-learning models provide much more flexible representations,

$$U_{\theta}(\mathbf{R}) = U_{\text{NN}}(\mathbf{R}; \theta), \quad (42)$$

while incorporating physical symmetries such as translational, rotational, and permutational invariance.

This increased flexibility can help represent the many-body interactions that naturally emerge from coarse graining.

However, increasing model flexibility does not change the fundamental statistical objective:

$$\boxed{p_{\theta}(\mathbf{R}) \approx p_{\text{target}}(\mathbf{R})}. \quad (43)$$